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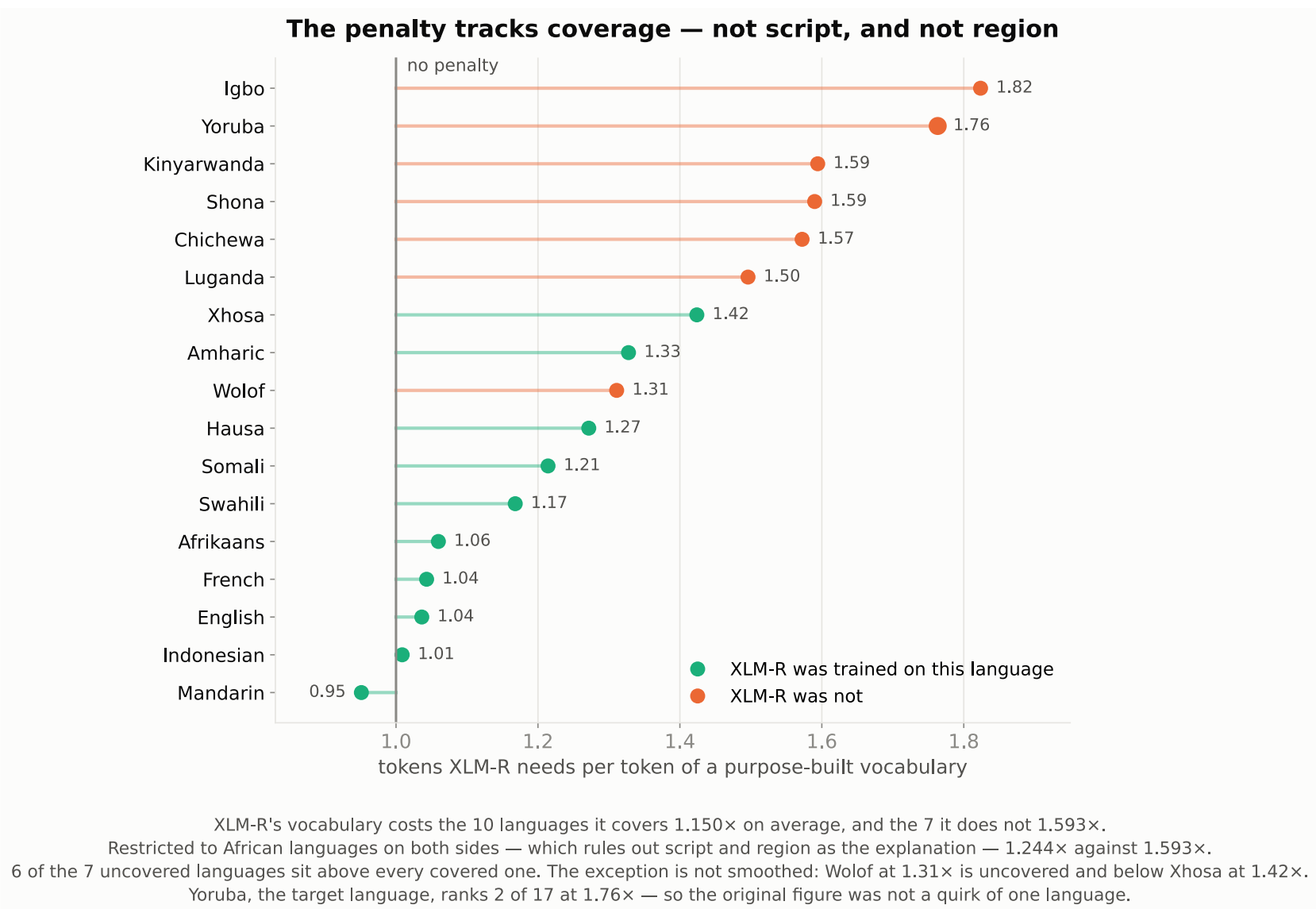
When Is It Worth Training Your Own Model?

Yoruba from scratch against multilingual transfer

A2-NLP Project Team · CSED 504 · University of Washington · Summer 2026

TAKEAWAY A 33.8M-parameter Yoruba model is ahead of mmBERT on topic classification; the strongest evidence points to vocabulary fit—not text scarcity—as the leverage.

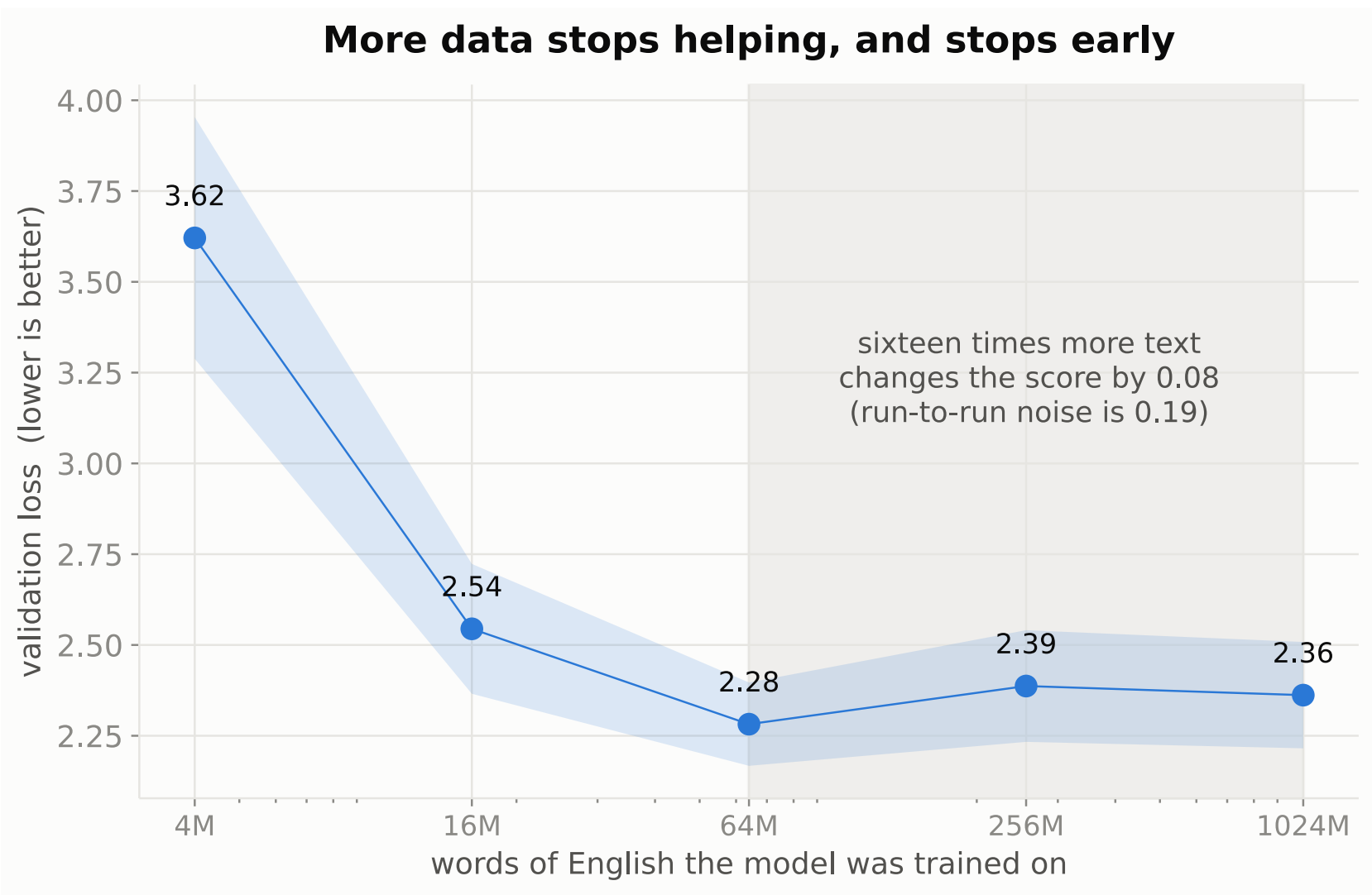
WHY VOCABULARY FIT MATTERS



Enough text

FineWeb-2 contains 69.1M Yoruba tokens. At fixed compute, an English control gained nothing measurable from 64M to 1,024M tokens, and Yoruba's 16M→64M change also sat inside seed variation. At these budgets, lack of additional text does not explain the result.

FIXED-COMPUTE CONTROL



THE EVIDENCE CHAIN

0.688

SIB-200 macro-F1 · from scratch

1.76×

XLM-R tokens per fitted token

69.1M

available Yoruba training tokens

33.8M

parameters in the small encoder

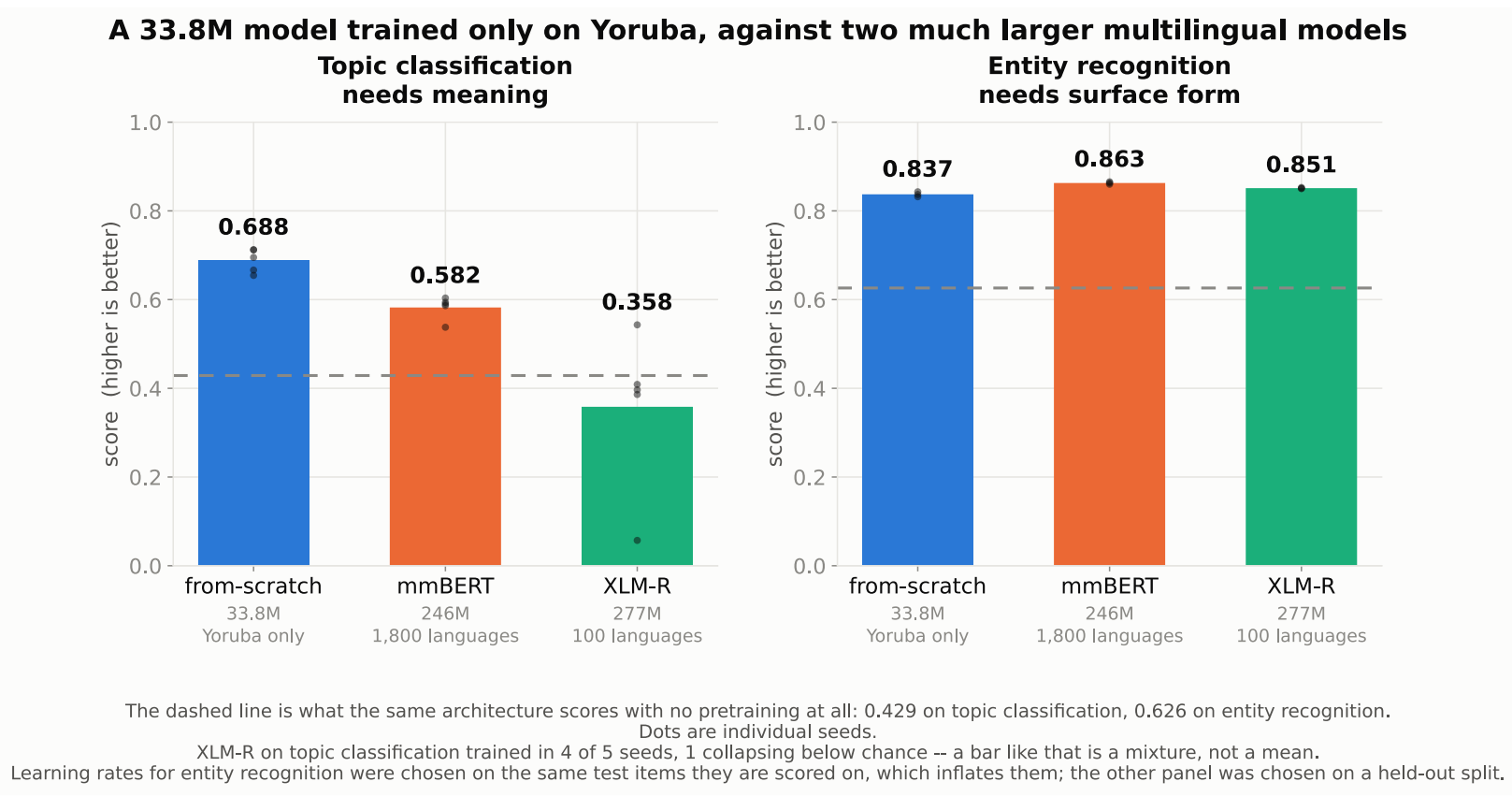
MEAN
No reliable matched-compute difference in pretraining bits per character (Welch $p=0.374$).

VARIANCE
The 250k-vocabulary arm was nearly four times as variable (sd 0.145 vs 0.037; $p=0.0098$).

DOWNSTREAM
Holding compute fixed, the fitted vocabulary led by +0.144 on SIB-200 and +0.061 on NER; every seed won.

NEXT TEST
Repeat the matched-compute vocabulary swap across the coverage gradient. Causal downstream evidence is still Yoruba-only.

DOWNSTREAM RESULT



Tasks disagree

MasakhaNER tells a different story: from scratch scored 0.837, behind mmBERT at 0.863. An untrained encoder still reached 0.626, showing how much entity recognition can obtain from capitalization and name shape. NER learning rates were selected on the scored items, so these values are descriptive, not a held-out model-selection comparison.

VARIANCE, NOT MEAN

